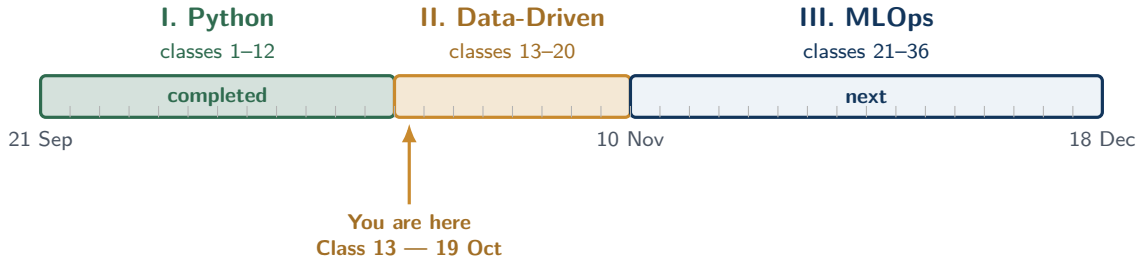
Class 13: Data Types and Structures for Data-Driven Systems

Data-driven Systems Engineering

Where we are in the course



Done: the tool

Syntax, data structures, functions, OOP, pandas, plotting, first EDA, scikit-learn API.

Now: the data

Data types, encoding, feature engineering, modeling, online ML, serving.

Next: the system

Software engineering, requirements, agile, MLOps, CI/CD, monitoring, project.

Agenda

- Why identifying data type comes before any modeling decision.
- Tabular data: samples, features, and the independent and identically distributed (i.i.d.) assumption.
- Time series data: order, timestamps, and temporal dependency.
- Spectral and signal data: waveforms and frequency-domain thinking.
- Image data: pixels, grids, and spatial structure.
- How the fraud and pasteurization cases embody different data types.

Learning goals

Learning goals

- Recognize that "a dataset" is not one thing: tabular, time series, spectral, and image data carry different assumptions.
- Identify which assumptions (independence, ordering, locality) each data type licenses or forbids.
- Connect each data type to a concrete artifact already in this repository.
- Explain why data type must be settled before encoding, feature engineering, or model choice.

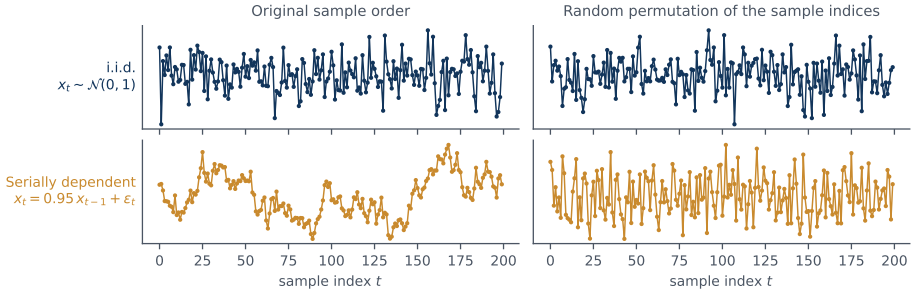
Why data type matters before modeling

- Every learning algorithm carries an inductive bias, i.e. structural assumptions about the data: exchangeability of samples, serial dependence over time, or locality over a spatial domain.
- Applying a method whose assumptions are violated by the data (e.g. an i.i.d. method on serially dependent observations) discards the very structure that carries the information.
- Data type is a property of the data-generating process, not of the storage format: the same physical quantity (temperature) may constitute i.i.d. samples in one study and a time series in another.
- Fixing the data type therefore constrains the admissible representations, feature constructions, and model families.

The i.i.d. assumption

Independent: $p(\mathbf{x}_i, \mathbf{x}_j) = p(\mathbf{x}_i)\, p(\mathbf{x}_j)$ for $i \neq j$; observing one sample conveys no information about another.

Identically distributed: every sample is drawn from the same $p(\mathbf{x}, y)$; the generating process does not change across samples.

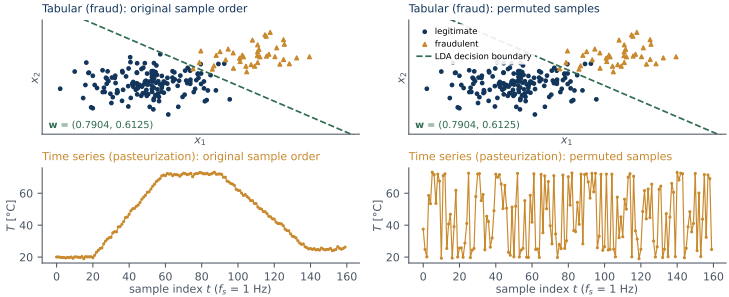


Permutation test: under the i.i.d. assumption, a random permutation of the sample indices leaves the joint distribution unchanged (top); under serial dependence, the same permutation destroys the temporal structure (bottom).

Tabular data versus time series under permutation

Tabular: the dataset is a multiset $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$; estimators that depend only on the empirical distribution are invariant to permutations of the sample indices.

Time series: the dataset is a sequence $(x_1, \ldots, x_T)$; the order is part of the data, and quantities such as $\dot{T}(t)$ depend on it.



Top: the estimated linear discriminant $\mathbf{w}$ is identical before and after the permutation. Bottom: permuting the sensor record eliminates the heating, holding, and cooling phases.

Notation

- A supervised dataset is a collection of n samples (instances, observations), possibly with repetitions:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}, \qquad \mathbf{x}_i = (x_{i,1}, \ldots, x_{i,d})^\top \in \mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_d, \quad y_i \in \mathcal{Y}.$$

- $x_{i,j} \in \mathcal{X}_j$ is the value of feature (attribute, variable) $j \in \{1, \ldots, d\}$ for sample i, where $\mathcal{X}_j$ is the domain of feature j (e.g. $\mathbb{R}$ or a finite set of categories); y_i is the corresponding target.
- Once all features are numerically encoded (Class 14), $\mathcal{X} \subseteq \mathbb{R}^d$ and the samples form the design matrix $X \in \mathbb{R}^{n \times d}$, whose i-th row is $\mathbf{x}_i^\top$ and whose j-th column collects feature j.
- Features are classified by measurement scale (Stevens, 1946):
  - **nominal** (categories without order), **ordinal** (ordered categories),
  - **interval** and **ratio** (numerical; continuous or discrete).

Single-target and multi-target problems

With $q > 1$ targets per sample, $\mathbf{y}_i = (y_{i1}, \ldots, y_{iq}) \in \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_q$, and the targets form an $n \times q$ matrix Y.

Problem	Target space	Example
Binary classification	$y_i \in \{0, 1\}$	Fraud detection (`is_fraud`)
Multi-class classification	$y_i \in \{1, \ldots, K\}$	Process phase s_t, $K = 6$
Multi-label classification	$\mathbf{y}_i \in \{0, 1\}^q$	Concurrent alarms in a batch
Multi-target regression	$\mathbf{y}_i \in \mathbb{R}^q$	Forecast of $T_{t+1}, \ldots, T_{t+h}$

- *Multi-output* and *multi-target* are synonyms; multi-label classification is the special case $\mathcal{Y}_k = \{0, 1\}$.
- Central issue: dependence among targets. Per-target models (binary relevance) ignore it; joint models (classifier chains, multivariate regression) exploit it.
- Evaluation requires dedicated metrics, e.g. Hamming loss and subset accuracy.

Tabular data

- Tabular data is represented by a design matrix $X \in \mathbb{R}^{n \times d}$ with a fixed, semantically meaningful feature set shared by all samples.
- The standard assumption is that samples are independent and identically distributed:

$$(\mathbf{x}_i, y_i) \overset{\text{i.i.d.}}{\sim} p(\mathbf{x}, y), \qquad i = 1, \ldots, n.$$

- Consequence: the empirical risk is invariant to any permutation π of the sample indices, so the order of samples carries no information.
- Features may be heterogeneous in scale (nominal, ordinal, numerical), which motivates the encoding techniques of Class 14.
- Most classical learning methods (linear models, decision-tree ensembles, k-NN, SVMs) are formulated under this assumption.

Tabular example: the fraud dataset

$n = 284{,}807$ samples

transaction_time	V1_feature	V2_feature	⋯	V28_feature	transaction_amount	is_fraud
$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	⋯	$x_{1,29}$	$x_{1,30}$	y_1
$x_{2,1}$	$x_{2,2}$	$x_{2,3}$	⋯	$x_{2,29}$	$x_{2,30}$	y_2
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$x_{n,1}$	$x_{n,2}$	$x_{n,3}$	⋯	$x_{n,29}$	$x_{n,30}$	y_n

features: $\mathbf{x}_i \in \mathbb{R}^{30}$

target: $y_i \in \{0, 1\}$

Feature	Scale	Description	Summary
`transaction_time`	interval	Seconds elapsed since the first transaction	0 – 172,792 s (≈ 48 h)
`V1_feature, ..., V28_feature`	interval	Principal components of anonymized attributes	mean ≈ 0; s.d. 1.96 (V1) to 0.33 (V28)
`transaction_amount`	ratio	Monetary value of the transaction	median 22.00; max 25,691.16
`is_fraud`	nominal	Target: 1 if fraudulent, 0 otherwise	prevalence 0.173%

Source: `Credit_Card_Fraud_Detection_Solution.ipynb`, output of `df.describe()`.

A taxonomy of data types

Data type	Index domain	Dependence structure	Expected invariance	Typical model families
Tabular	None (sample set)	Independent samples	Permutation of samples	Linear models, tree ensembles, k-NN
Time series	Time $t \in \{1, \dots, T\}$	Serial dependence, $\rho(k) \neq 0$	Shift in time (if stationary)	ARIMA, state-space models, RNNs
Signal / spectrum	Time or frequency	Periodic components	Shift in time; phase	DFT/STFT features, 1D CNNs
Image	Spatial grid (u, v)	Local spatial correlation	Translation	CNNs, vision transformers

- The invariance column states which transformations of the input should leave the prediction unchanged; it is what a well-chosen model family encodes by construction.

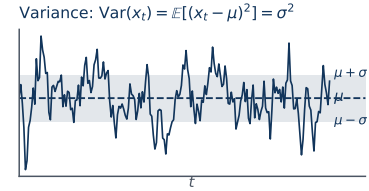
Time series data

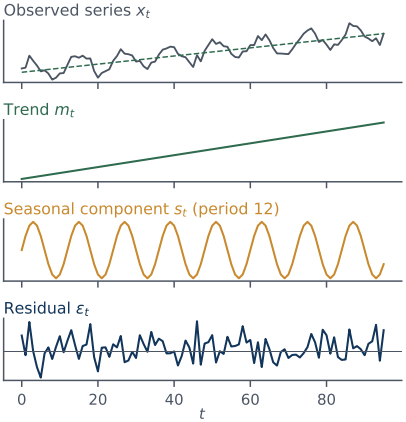
- A time series is a realization $\{x_t\}_{t=1}^{T}$ of a stochastic process indexed by time; the index is part of the data, not a storage artifact.
- Observations exhibit serial dependence, quantified by the autocorrelation function

$$\rho(k) = \frac{\mathrm{Cov}(x_t, x_{t-k})}{\mathrm{Var}(x_t)}, \qquad \rho(k) \neq 0 \text{ for small lags } k.$$

- A process is (weakly) stationary if its mean and autocovariance do not depend on t; trend, seasonality, and regime changes violate this.
- Classical decomposition: $x_t = m_t + s_t + \varepsilon_t$ (trend, seasonal component, residual).
- The i.i.d. assumption does not hold, so random resampling for model validation is not admissible.

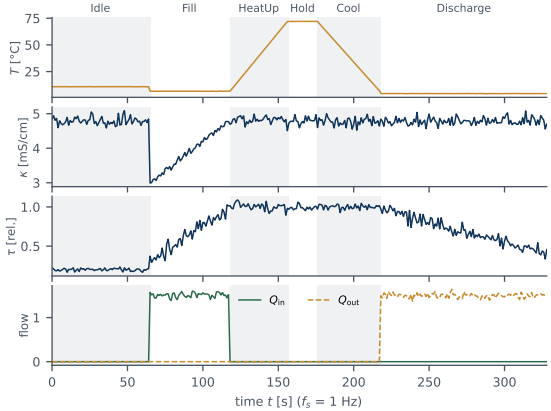
Variance, covariance, and decomposition





Left: a stationary process, with $\rho(1) = \mathrm{Cov}(x_t, x_{t-1})/\sigma^2$. Right: additive decomposition $x_t = m_t + s_t + \varepsilon_t$.

Time series example: a pasteurization batch



One batch from `synth_sensors.py`; $\mathbf{x}_t \in \mathbb{R}^8$, four of the eight channels shown.

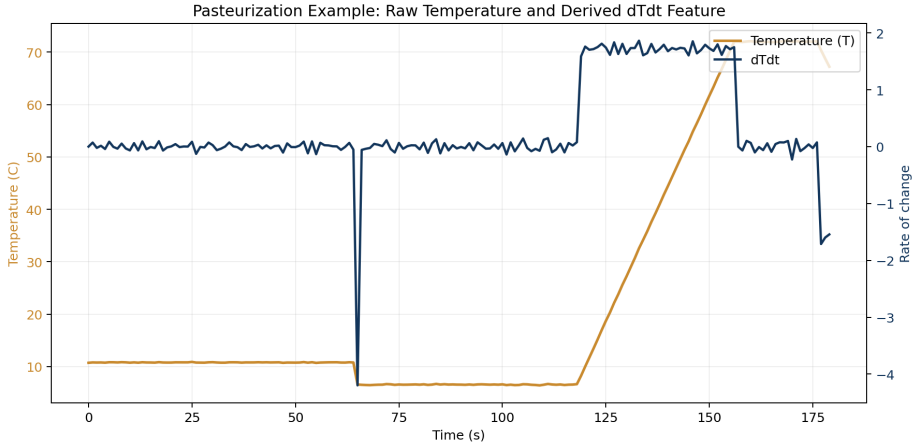
Task

Given the window $W_t = (\mathbf{x}_{t-w+1}, \ldots, \mathbf{x}_t)$, predict the process phase

$$s_t \in \mathcal{S} = \{\text{Idle}, \ldots, \text{Discharge}\}.$$

1. Classify the task by its target space.
2. Identify the channels that discriminate Fill from Discharge.
3. Explain why $\dot{T}(t) \approx (T_t - T_{t-1})/\Delta t$ is informative for HeatUp and Cool.
4. Propose a train/test split that is valid for this dataset.

From raw signal to derived feature

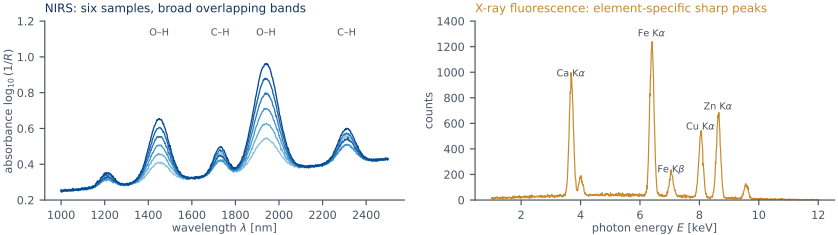


Temperature T_t and its finite-difference derivative $\dot{T}(t)$ from `pasteurization/synth_sensors.py`: the derived feature makes the heating and cooling regimes explicit.

Spectral and signal data

- A discrete signal $x[n] = x(n/f_s)$ is obtained by sampling a continuous phenomenon at frequency f_s (audio, vibration, high-rate sensors).
- Nyquist–Shannon: a component of frequency f is recoverable only if $f_s > 2f_{\max}$; otherwise it is aliased onto lower frequencies.
- The discrete Fourier transform (DFT) re-expresses a length-N signal as a combination of N frequency components.
- Non-stationary signals are analysed with the short-time Fourier transform (windowed DFT); in speech, MFCCs summarize each window on a perceptual frequency scale.
- The repository contains no audio dataset; the pasteurization temperature trace serves as the reference example of a sampled physical signal.

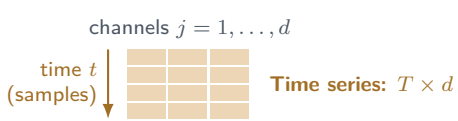
Spectroscopic data: NIRS and X-ray



- Each sample is a spectrum $\mathbf{x}_i = (x_i(\lambda_1), \ldots, x_i(\lambda_d))$ indexed by wavelength or energy, not by time; adjacent variables are strongly correlated and typically $d \gg n$.
- **Near-infrared spectroscopy (NIRS)**, 780–2500 nm: broad, overlapping overtone and combination bands of O–H, C–H, and N–H bonds; used to estimate composition (moisture, fat, protein). Standard pipeline: scatter correction (SNV, MSC), Savitzky–Golay derivatives, partial least squares (PLS) regression.
- **X-ray spectroscopy**: fluorescence (XRF) yields sharp peaks at element-specific energies (elemental composition); diffraction (XRD) yields peaks over the angle 2θ (crystalline phases). X-ray radiographs, by contrast, are images.

Illustrative synthetic spectra; band and line positions follow standard reference values.

Time series versus spectral data



	Time series	Spectral data (e.g. NIRS)
Index	Time t; indexes the samples	Wavelength λ or energy E; indexes the features (spectra are usually i.i.d.)
Direction	Causal: the past determines the future	Non-causal: no preferred direction, two-sided filters are admissible
Typical question	Forecasting x_{t+h}, detecting regime changes	Predicting a property y_i from the whole spectrum
Valid split	Temporal (train before test)	By physical sample (replicates kept together)

- **Common ground:** both are functions sampled on an ordered grid, with strong correlation between neighbouring points.
- **Relation:** the Fourier transform maps a time series to its frequency spectrum; FT-NIR instruments obtain spectra by Fourier-transforming a measured interferogram.

Image data

- An image is a tensor $I \in \mathbb{R}^{H \times W \times C}$ ($C = 1$ for grayscale, $C = 3$ for red–green–blue (RGB) channels), i.e. a function on a discrete spatial grid (u, v).
- Spatial locality: correlation between pixels decays with their distance, so relevant patterns are local.
- Semantic content is (approximately) invariant to translation; convolution is translation-equivariant, $f(\tau_\delta I) = \tau_\delta f(I)$, which motivates convolutional neural networks (CNNs).
- The index domain is two-dimensional and has no natural total order: flattening to a vector of length $H \cdot W \cdot C$ discards the neighborhood structure.
- The repository contains no image dataset; the type is included to complete the taxonomy before representation is discussed in Class 14.

Consequences of ignoring the data type

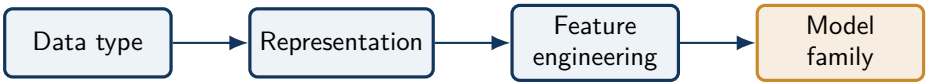
Data type	Assumption violated by a naive tabular treatment
Time series	Exchangeability: random splits place future samples in the training set (temporal leakage).
Signal / spectrum	Representation: raw amplitudes in the time domain obscure the periodic components, which are sparse in the frequency domain.
Image	Locality: flattening to $\mathbb{R}^{H \cdot W \cdot C}$ discards spatial neighborhoods and translation structure.

Data leakage and valid evaluation

Definition: leakage occurs when information that would be unavailable at prediction time influences model training or model selection, yielding optimistically biased performance estimates.

Type	Mechanism	Example	Remedy
Target leakage	A feature encodes the target or is recorded after it	A chargeback flag used to predict `is_fraud`	Retain only features available at prediction time
Preprocessing contamination	Transformations fitted before splitting	Scaling, imputation, PCA, or oversampling fitted on all data	Fit transformations on the training fold only (pipeline)
Temporal leakage	Future observations inform the past	Random k-fold on the sensor log; centred rolling means	$\max t_{\text{train}} < \min t_{\text{test}}$; rolling origin; causal features
Group leakage	Samples of one entity on both sides of the split	Windows of one batch; replicate spectra of one sample	Group-wise split
Duplicate leakage	(Near-)identical samples in train and test	Repeated transactions or re-measured spectra	Deduplicate before splitting
Model-selection leakage	The test set guides tuning or feature selection	Hyperparameters chosen by test-set error	Validation set or nested cross-validation

From data type to system design



- Each stage is the subject of a subsequent class: representation and encoding (Class 14), feature engineering (Class 16), and modeling (Class 18).
- An error at the first stage propagates to every subsequent stage of the pipeline.

Repository connections

Repo activity

- `Credit_Card_Fraud_Detection_Solution.ipynb` and `streamlit/modeling.py`: tabular data, one sample per transaction.
- `pasteurization/synth_sensors.py`: multivariate time series, eight sensor channels sampled at 1 Hz over a batch lifecycle.
- `pasteurization/serving.py`: the same process delivered as a data stream, used in the practical session of Class 20.

Common methodological errors

- Treating a time-ordered sensor log as an i.i.d. sample and randomly permuting it before the train/test split.
- Interpreting feature-importance measures computed on raw pixels as if pixels were semantically meaningful features.
- Sampling below the Nyquist rate, so that the frequencies of interest are aliased and irrecoverable.
- Equating sample size with information: under serial dependence the effective sample size is smaller than n.

Papers and materials

- Stevens, S. S. (1946). On the theory of scales of measurement. *Science*, 103(2684), 677–680.
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- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley.
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- Kapoor, S., Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804.
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Take-away message

Identify the data type before any modeling decision:
it determines the representation,
the admissible models, and the validation protocol.

Thank you for your attention.

Next: Class 14 — structured, semi-structured, and unstructured data; encoding and embedding.